

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

- For water: Use the formula $Q = m * c * \Delta T$ where $m = 1000\text{ g}$ (1 liter of water), $c = 1\text{ cal/g}^\circ\text{C}$, $\Delta T = 60^\circ\text{C} - 30^\circ\text{C} = 30^\circ\text{C}$.

- For aluminum: Use the same formula $Q = m * c * \Delta T$ where $m = 1000\text{ g}$ (1 kg), $c = 0.215\text{ cal/g}^\circ\text{C}$, $\Delta T = 60^\circ\text{C} - 30^\circ\text{C} = 30^\circ\text{C}$.

- Calculate Q for water: $Q = 1000 * 1 * 30 = 30000\text{ cal} = 30\text{ Kcal}$.

- Calculate Q for aluminum: $Q = 1000 * 0.215 * 30 = 6450\text{ cal} = 6.45\text{ Kcal}$.

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<answer> 30 Kcal, 6.45 Kcal </answer>